

BENNETT JACKSON

SOFTWARE ENGINEER

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EDUCATION

University of Arizona | B.S. in Software Engineering (May 2025)

Aug 2021 - Present

GPA | 3.84

Awards | Dean's List with distinction (x3), Academic year distinction (x2), Engineering Honors Convocation Award (x2)

Extracurricular | Student advisory board for Software Engineering, Member of Software Engineering Wildcats Club

EXPERIENCE & PROJECTS

Student Software Engineer | University of Arizona | *Tucson, AZ*

Jan 2024 - May 2024, Aug 2024 - Present

- Developed a full-stack web application for the Systems and Industrial Engineering department to convert and visualize Bayesian networks from OML, including a parser for efficient data translation, enhancing accessibility for researchers.
- Conducted thorough testing and debugging, achieving a 95% reduction in reported bugs during the testing phase.
- Built a Flask backend with six RESTful API endpoints and containerized the app using Docker for seamless deployment.
- Created a responsive frontend using HTML, CSS, and JavaScript, enhancing usability and satisfaction by 50%.

Software Engineer Intern | Garmin | *Chandler, AZ*

May 2024 - Aug 2024

- Developed safety-critical embedded C software for several generations of Garmin Integrated Flight Decks, ensuring compliance with industry standards across 80+ aircraft models.
- Enhanced the testing process for 15 components by developing an automated regression testing suite in Python.
- Identified and resolved seven bugs found during testing through software and requirement changes.
- Implemented six customer-requested airframe features to enhance the functionality of Garmin Integrated Flight Decks.
- Leveraged software architecture to introduce runtime-determined features, boosting system flexibility and adaptability.
- Utilized Git to manage and track changes, pushing 75+ commits to master during software development.

Software Engineer Intern | NASA Jet Propulsion Laboratory | *Pasadena, CA*

Jan 2023 - Jan 2024

- Streamlined Python scripts with open-source libraries, Jama, and TestRail APIs, improving runtime by 32%.
- Led a team of three interns to develop an open-source software solution for an internal application migration.
- Implemented CI/CD pipelines with Jenkins for automated testing and deployment of two pip packages.
- Effectively managed tickets and metrics across multiple sprints using Jira, successfully completing 90+ tickets.
- Spearheaded internal customer-facing interactions to communicate project updates and address concerns.
- Utilized Git and GitHub for version control and collaboration, successfully merging 90+ pull requests to master.

Project | Weekly Playlist Manager

- Constructed in Python, this project leverages the Spotify API to automate playlist management, allowing users to retrieve and save songs seamlessly. Integrated CI/CD with GitHub Actions for regular updates and efficient deployment.

KEY SKILLS AND RELEVANT COURSEWORK

Coding Languages | C, C++, Python, Java, HTML, CSS, JavaScript, Verilog, Assembly, Bash, Swift

Technical Skills | Linux, Unix, Command-Line Interface, Git, GitHub, CI/CD, Jenkins, Docker, Maven, GitHub Actions, Jira, SQL, Spring, Spring Boot, RESTful APIs, JUnit, Unit Testing, Software Security, Data Engineering, ETL, UML, OML, Flask, Pandas, Jama, Scrum and Agile Methodologies, Object-Oriented Programming, Open-source Practices, Conda

Additional Skills | Creative Solutions, Strategic Thinking, Teamwork, Communication, Software Development Lifecycle

Coursework | Software Architecture and Design, Software DevSecOps, Introduction to Cybersecurity Operations, Software Assurance and Security, Software Requirements Analysis, Software Project Management, Discrete Structures and Basic Algorithms, Computer Programming for Engineering Applications 1 and 2, Systems Cybersecurity Engineering, Fundamentals of Computer Organization, Discrete Math, Digital Logic, Interdisciplinary Capstone, Introduction to Web Design and Development, Linear Algebra, Engineering Ethics